

**SMGT 530 Final Paper**

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## Introduction

Competitive games create a natural problem for data analysis: players want a clear way to understand “who is strong” and “who counters whom,” but the underlying game is noisy, strategic, and constantly changing. League of Legends is a perfect example. Champions do not have fixed “strength tiers” that can be directly observed; instead, we only see outcomes from many games played under different conditions. This paper mainly discusses whether we can turn those noisy outcomes into an interpretable, statistically grounded ranking.

My first research question is: Can a Bradley–Terry (BT) paired-comparison model recover a meaningful tier ranking for League of Legends top-lane champions from matchup data? In other words, if we treat each champion-versus-champion pairing as a head-to-head contest and use the observed win rates across many games, can the BT model estimate latent “strength” parameters that produce a sensible ordering (and therefore tier list) of top-lane champions? This is appealing because the BT model is designed exactly for turning paired outcomes into an interpretable scale of relative ability.

My second research question builds on the first: Beyond an overall tier ranking, can the dataset reveal interactive matchup effects between champions? A single global strength score assumes a transitive world where stronger champions tend to beat weaker ones consistently. But real gameplay often contains “rock–paper–scissors” structure: a champion may be average overall but unusually strong or weak into particular opponents due to kit interactions, laning patterns, range advantage, sustain, crowd control, or scaling. If such champion-specific interactions exist in the data, I want to identify and quantify them in a way that is still explainable.

## Background

Riot Games does not publish an official, explicit champion strength list. Instead, Riot provides an API that allows third-party websites to collect match data and produce their own tier lists, rankings, and “weak against” suggestions. These rankings strongly influence player perception and decision-making (for example, which champions are considered meta, which picks are considered safe blind picks, and which matchups are seen as counters). However, a key limitation is that most third-party sites do not fully disclose the algorithms behind their tier lists or matchup recommendations. From a player’s perspective, the rankings often behave like a

black box: the output is easy to consume, but the reasoning is not transparent, and it is difficult to evaluate why a champion is ranked where it is or why a particular “weak against” label appears.

Lane choice also matters because role dynamics can introduce confounding that makes “individual champion strength” harder to isolate. I focus on top lane because it provides the cleanest environment for champion-versus-champion signal. Top lane is typically a solo lane with frequent direct interactions over long laning phases, and it is less inherently dependent on synergy with a lane partner. This helps reduce confounds and makes matchup outcomes more interpretable as evidence of relative champion strength in that lane.

By contrast, other roles introduce additional structure that can blur the meaning of a one-champion strength estimate. Jungle impact is often expressed through ganks, vision control, and objective pressure, meaning outcomes depend heavily on cross-lane decisions and team coordination rather than sustained, isolated “lane fights.” Mid lane is shorter and often emphasizes wave control and map influence; roaming and coordination can dominate pure 1v1 laning, making it harder to attribute outcomes cleanly to champion matchups. Bot lane is fundamentally a duo lane where champion pairing and support synergy matter substantially (for example, combinations like Lucian with Nami), so a single-champion rating can be misleading unless the model explicitly accounts for pair effects. For these reasons, top lane is a reasonable starting point for testing whether a paired-comparison model can build an explainable tier list and whether matchup-specific interaction effects can be detected reliably.

### **Data**

The data for this project comes from OP.GG, which is one of the most widely used League of Legends analytics websites. It provides aggregated statistics such as matchup win rates and game counts, and it is commonly trusted by players as a practical reference for champion strength. Because my goal is to estimate champion strength from head-to-head outcomes, I treat each champion-versus-champion matchup as a summarized record that includes how often the matchup occurred and how often the champion won.

To reduce bias and make the signal cleaner, I limit the scope in two important ways. First, I only use top-lane matchups. Compared with other roles, top lane is typically a solo lane with more consistent direct champion interactions, so the outcome is less dominated by duo synergy or constant cross-map involvement. Second, I only use games from Master rank and above. League of Legends ranked distribution is pyramid-shaped, meaning most games happen at lower

ranks. At those ranks, outcomes can be heavily influenced by player mistakes, inconsistent execution, and differences in champion familiarity, which makes it harder to interpret win rates as reflecting champion strength rather than player skill effects. By focusing on Master+, the dataset reflects games where players are generally more consistent, so the matchup outcomes are closer to measuring the champions themselves.

I collected the data using a Python web scraper that I wrote for this project. The scraper visits OP.GG pages for top-lane champions and extracts matchup-level statistics. The output is a dataset of champion–opponent pairs with their observed win rate and the number of games behind that win rate, which provides both performance and reliability information for each matchup..

## **Methods**

### **Tier ranking using an explainable paired-comparison model**

To build a transparent tier ranking, I use a paired-comparison approach from class: the Bradley–Terry model. Conceptually, the model assumes every champion has an underlying strength value, and when two champions face each other, the chance of winning depends on which champion’s strength is higher. The model learns these strength values by finding the set of champion strengths that best explains the matchup win results across the entire dataset. Once the strengths are estimated, I can rank champions from strongest to weakest and convert that ordering into tiers.

### **Regularization to improve stability**

When I first applied the raw Bradley–Terry model, the ranking was not stable enough to be convincing. This can happen because some matchups have small sample sizes, some champions have uneven matchup coverage, and a few extreme win rates can overly influence the estimated strengths. To address this, I add regularization, which is a standard technique that gently pulls champion strength estimates away from extreme values unless the data provides strong evidence.

### **Finding special counters with a statistical test**

To identify “special counters,” I test whether a champion’s performance against a specific opponent is unusually different from that champion’s typical performance across all matchups. The null hypothesis is that the champion’s chance of winning in that particular matchup is the same as their overall chance of winning. The alternative hypothesis is that the

matchup chance of winning is different from their overall level, meaning the opponent is a meaningful counter (either favorable or unfavorable).

I use a standard Z-test style approach to measure how far the matchup win rate deviates from the champion's overall win rate, taking into account how many games were observed in that matchup. Matchups supported by more games are treated as more reliable, while matchups with very few games are less likely to be flagged. I use a threshold corresponding to a 95% confidence level, meaning I only label a matchup as a special counter when the deviation is large enough that it is unlikely to be explained by random sampling variation alone. Positive deviations indicate the champion performs unusually well into that opponent, while negative deviations indicate the champion performs unusually poorly.

### **Result**

The raw Bradley–Terry tier list (Image 1) produced a ranking that looked highly biased toward low pick-rate champions. The strongest champions near the top of the list include several champions that are commonly used as counter picks rather than as broadly “best-in-role” blind picks. This makes sense in practice: many low pick-rate top-lane champions are selected specifically when the player already sees a favorable opponent, so their matchup outcomes are not coming from a neutral sample of games. Because the raw BT model tries to explain win/loss outcomes purely through a single strength value per champion, it can mistakenly interpret “often picked only when favorable” as “universally strong.” In other words, the model amplifies the counter-pick advantage and turns it into an inflated global strength estimate, which leads to an unrealistic tier list compared to what players expect from the meta.

To improve the stability and realism of the ranking, I first filtered out very low pick-rate champions using a 2% pick-rate threshold, and then fit a regularized version of the BT model (Image 2). This second ranking was noticeably more reasonable. The estimated strengths were less extreme, and the ordering aligned much better with OP.GG's published top-lane tier list. When I compared the regularized tier list to OP.GG's list, there was substantial overlap in the champions that appeared near the top and the general grouping of strong versus weak champions. This suggests that the combination of filtering (to reduce counter-pick selection bias) and regularization (to reduce overreaction to noisy matchups) helped the BT approach recover a tier ranking that better matches community expectations and the widely used “black box” reference ranking.

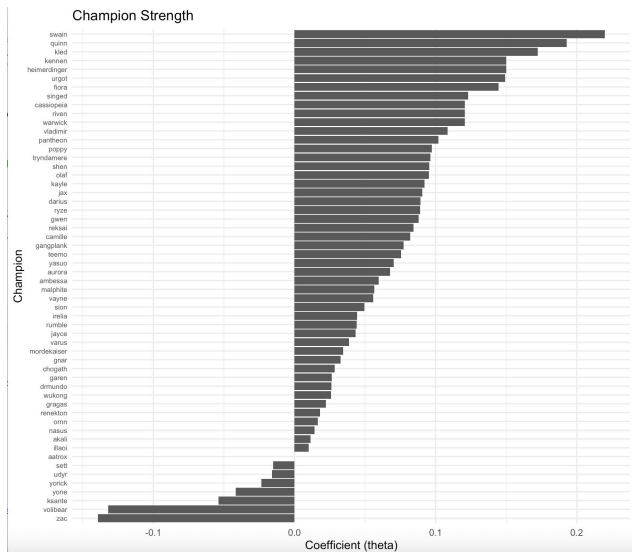


Image 1

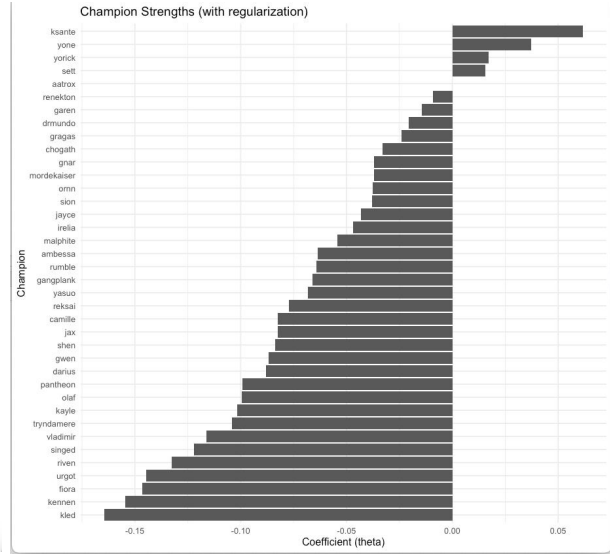


Image 2

For matchup-specific “special counters,” the hypothesis-testing results (Image 3) identified many champion–opponent pairs where the observed matchup win rate deviated strongly from the champion’s typical performance. The results show clear and interpretable counter patterns rather than random noise: certain champions repeatedly appear as unusually good into specific opponents, and certain matchups repeatedly show unusually poor performance. This is exactly the behavior players mean when they talk about counters, and it provides an explainable, statistics-based alternative to OP.GG’s “weak against” tags.

|    | champion    | opponent    | n_games | p_obs  | p_exp     | z_score  | p_value      | lane_pick_rate |
|----|-------------|-------------|---------|--------|-----------|----------|--------------|----------------|
| 1  | mordekaiser | volibear    | 79      | 0.6962 | 0.4961655 | 3.555995 | 0.0003765505 | 0.0422         |
| 2  | fiора       | ksante      | 384     | 0.6084 | 0.5227494 | 3.360284 | 0.0007786243 | 0.0516         |
| 3  | yone        | ksante      | 234     | 0.5823 | 0.4757318 | 3.264206 | 0.0010977115 | 0.0401         |
| 4  | sion        | yorick      | 139     | 0.6312 | 0.4981378 | 3.137582 | 0.0017034756 | 0.0764         |
| 5  | jax         | darius      | 290     | 0.5972 | 0.5084958 | 3.021593 | 0.0025144826 | 0.0686         |
| 6  | kayle       | ksante      | 221     | 0.6091 | 0.5106455 | 2.927927 | 0.0034123040 | 0.0246         |
| 7  | cassiopeia  | mordekaiser | 37      | 0.7568 | 0.5192555 | 2.891999 | 0.0038279862 | 0.0150         |
| 8  | akali       | yone        | 84      | 0.6429 | 0.4880425 | 2.839397 | 0.0045198933 | 0.0194         |
| 9  | aatrox      | warwick     | 98      | 0.6263 | 0.4855478 | 2.787917 | 0.0053048212 | 0.0918         |
| 10 | ambessa     | ksante      | 678     | 0.5533 | 0.5015003 | 2.697580 | 0.0069845582 | 0.1003         |
| 11 | gragas      | gnar        | 76      | 0.6447 | 0.4908434 | 2.683032 | 0.0072958032 | 0.0313         |
| 12 | yorick      | jayce       | 173     | 0.5799 | 0.4784260 | 2.671854 | 0.0075433548 | 0.0232         |
| 13 | shen        | ksante      | 246     | 0.5935 | 0.5098296 | 2.625145 | 0.0086612174 | 0.0394         |
| 14 | warwick     | mordekaiser | 35      | 0.7353 | 0.5161101 | 2.594837 | 0.0094635865 | 0.0151         |
| 15 | singed      | drumundo    | 91      | 0.6522 | 0.5165926 | 2.588649 | 0.0096353117 | 0.0291         |
| 16 | illaoi      | ornn        | 58      | 0.6607 | 0.4922179 | 2.566554 | 0.0102714691 | 0.0118         |
| 17 | ksante      | nasus       | 70      | 0.6232 | 0.4709393 | 2.552123 | 0.0107068587 | 0.0808         |
| 18 | swain       | malphite    | 31      | 0.7742 | 0.5465394 | 2.546175 | 0.0108910585 | 0.0082         |
| 19 | gangplank   | singed      | 117     | 0.6228 | 0.5053611 | 2.540738 | 0.0110618921 | 0.0365         |
| 20 | jax         | mordekaiser | 187     | 0.6011 | 0.5084958 | 2.533053 | 0.0113073834 | 0.0686         |

Image 3

When I repeated the special-counter analysis focusing specifically on low pick-rate champions (Image 4), the results looked even more practically useful. This is because the strategic value of low pick-rate champions is often not that they are strong in general, but that they provide sharp matchup advantages in draft. Filtering to low pick-rate champions and then identifying statistically significant favorable matchups directly supports ban/pick strategy: it highlights where niche picks are most likely being used for a specific advantage rather than general strength. In that sense, the low pick-rate counter table matches the real-world purpose of these picks and gives actionable information that a global tier list alone cannot provide.

|    | champion   | opponent    | n_games | p_obs  | p_exp     | z_score  | p_value     | lane_pick_rate |
|----|------------|-------------|---------|--------|-----------|----------|-------------|----------------|
| 1  | cassiopeia | mordekaiser | 37      | 0.7568 | 0.5192555 | 2.891999 | 0.003827986 | 0.0150         |
| 2  | akali      | yone        | 84      | 0.6429 | 0.4880425 | 2.839397 | 0.004519893 | 0.0194         |
| 3  | warwick    | mordekaiser | 35      | 0.7353 | 0.5161101 | 2.594837 | 0.009463586 | 0.0151         |
| 4  | illaoi     | ornn        | 58      | 0.6607 | 0.4922179 | 2.566554 | 0.010271469 | 0.0118         |
| 5  | swain      | malphite    | 31      | 0.7742 | 0.5465394 | 2.546175 | 0.010891059 | 0.0082         |
| 6  | vayne      | camille     | 51      | 0.6731 | 0.5002926 | 2.468184 | 0.013580041 | 0.0157         |
| 7  | aurora     | yone        | 33      | 0.7188 | 0.5046174 | 2.460875 | 0.013859845 | 0.0148         |
| 8  | poppy      | ornn        | 62      | 0.6667 | 0.5120603 | 2.435977 | 0.014851623 | 0.0170         |
| 9  | aurora     | ksante      | 112     | 0.6195 | 0.5046174 | 2.431710 | 0.015027746 | 0.0148         |
| 10 | teemo      | ambessa     | 127     | 0.6032 | 0.5057597 | 2.196338 | 0.028067779 | 0.0162         |
| 11 | aurora     | darius      | 46      | 0.6596 | 0.5046174 | 2.102376 | 0.035520376 | 0.0148         |
| 12 | nasus      | yasuo       | 56      | 0.6250 | 0.4873354 | 2.061037 | 0.039299517 | 0.0172         |

Image 4

## Discussion

The biggest limitation of the BT approach in this project is what it chooses to “see” as evidence. In its basic form, BT is built around win/loss outcomes, and it compresses each champion into a single strength number. That simplification is useful for interpretability, but it also throws away important information. A key example is matchup reliability: some matchups are supported by a very large number of games while others appear only a few hundred times. In my current implementation, this difference is not reflected strongly enough, so a low-sample matchup can end up influencing the learned strengths more than it should. This helps explain why the raw BT model was vulnerable to low pick-rate and counter-pick effects: champions that show up mainly in favorable, lower-volume situations can look artificially strong if the model does not properly down-weight or stabilize those matchups.

A second weakness is selection bias from the draft process. Matchups in real games are not randomly assigned. Players choose champions with information (especially in counter-pick situations), and that means the observed outcomes reflect both champion power and player

intention. Low pick-rate champions are often chosen only when the player expects a good matchup, so their win rates are not representative of “how strong they are into the whole field.” BT, by design, does not model the champion-selection mechanism, so it can confuse “picked only when favorable” with “strong overall.” My filtering step and regularization reduced the damage, but they do not fully solve the underlying selection problem.

These weaknesses also explain why OP.GG still appears to produce a better ranking list than my BT tier list, even after improvements. OP.GG likely uses additional signals beyond pure win/loss, such as pick rate and ban rate, lane kill rate, KDA, and Kill percentage. My BT model is intentionally simpler and more explainable, but that simplicity limits accuracy.

### **Conclusion**

To conclude, after filtering out very low pick-rate champions and adding regularization, the ranking became much more stable and overlapped substantially with OP.GG’s published tier list. This suggests the BT framework can produce a reasonable and interpretable tier ordering when extreme estimates and selection-driven noise are controlled. In addition, the hypothesis-testing step for “special counters” successfully identified strong and consistent matchup patterns. These counter results were even more useful when focusing on low pick-rate champions, because the competitive value of these champions is often draft-specific advantage rather than universal dominance.

From a sport management perspective, the impact is practical. Many second-tier or semi-professional teams do not have the budget to hire a dedicated coach or analyst, yet they still need data-driven guidance for ban/pick decisions. Today, their primary source is third-party tier lists and “weak against” labels that operate as black boxes. This project provides a transparent alternative: an explainable tier list and an explainable set of statistically supported counters derived directly from observed match outcomes. The answer to the research question is therefore useful not only for improving competitive draft preparation, but also for changing how fans interpret tier lists—shifting the conversation from “trust the website” to “here is the model logic and evidence behind the ranking.”